

OPERATIONAL AND SALES ANALYSIS OF A PACKAGED DRINKING WATER COMPANY

A Presentation for BDM Capstone Project



Submitted By

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ABOUT THE BUSINESS

DETAILS :

NAME : ABN SUN Foods and Beverages
(Brand: IRA Gold Packed Drinking Water)

OWNER : Mr. Akshay Nanote

LOCATION :

Near SIT Nagpur , Bidgaon, Nagpur

TIMING :

10 A.M to 6 P.M.

OPERATION:

Operates in Both B2B and B2C

Products :

500ml Plastic Box, 1000ml Plastic Box
500ml Corrugated Box, 1000ml Corrugated Box

fig 1 : The owner of IRA Gold



fig 2 : Me with the owner of IRA Gold





PROBLEMS STATEMENTS & OBJECTIVES

Problems :

01

Machine utilization remains significantly below installed capacity due to irregular electricity supply and infrastructure limitations, restricting production output.

02

A substantial portion of sales occurs on credit, resulting in delayed cash inflow and short-term cashflow constraints that affect operational efficiency.

Objectives :

01

To analyze machine utilization and production efficiency using operational data and identify key bottlenecks.

02

To evaluate the impact of credit sales on cashflow and profitability and propose data-driven operational improvements.

DATA OVERVIEW

- The analysis is based on primary data collected directly from the business, using daily production records, sales bills, excel sheets and payment logs.
- The dataset covers a three-month period from June 1 to August 31, 2025, capturing regular operating conditions

01

Production data -
machine capacity, actual output, utilization, production cost, revenue, gross profits and remarks.

02

Sales and credit data -
quantity sold, revenue, payment mode, credit days, pending amounts and remarks



SOURCE OF DATA AND DATA CLEANING METHODS

Data Consolidation:

Sales data from manual sale slips (two months) and Excel records (one month) were merged into a single dataset after verifying consistency across key fields.

Duplicate Handling:

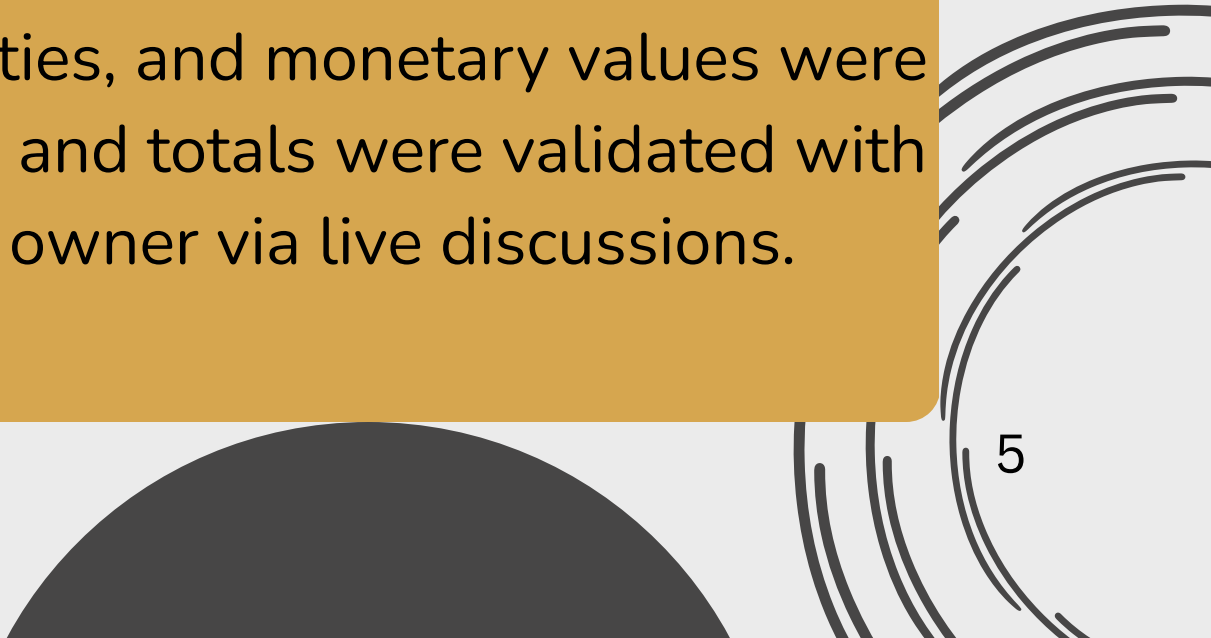
Overlapping entries across both sources were cross-checked to avoid duplication of transactions.

Handling Missing Values:

Missing entries were resolved by cross-verifying sale slips and Excel records. Only complete and verified transactions were retained for analysis.

Standardization & Validation:

Dates, quantities, and monetary values were standardized, and totals were validated with the shop owner via live discussions.



DESCRIPTIVE STATISTICS

fig 3 : descriptive stats daywise

		Day Wise		
Statistics	Actual Produ	Utilization (%)	Revenue (Rs)	Gross Profit (
Count:	92	92	92	92
Mean:	500.7282609	41.7275	60087.3913	10014.56522
Std. Dev :	11.00260035	0.916698118	1320.312042	220.0520071
Min :	481	40.08	57720	9620
25%:	491.75	40.98	59010	9835
50% Median:	500	41.67	60000	10000
75%:	511	42.58	61320	10220
Max:	520	43.33	62400	10400

fig 4: descriptive stats monthwise

		Month Wise		
Statistics	Actual Produ	Utilization (%)	Revenue (Rs)	Gross Profit
Count:	3	3	3	3
Mean:	500.774552	41.73333333	1842680	307113.3333
Std. Dev :	4.025465294	0.337095437	22099.66516	206107.6667
Min :	497.0322581	41.42	1818120	205758.7778
25%:	498.6451613	41.555	1833540	171074.6667
50% Median:	500.2580645	41.69	1848960	239659.9259
75%:	502.6456989	41.89	1854960	194313.7037
Max:	505.0333333	42.09	1860960	205497.7901

OBSERVATIONS :

- **Operational Stability:** Mean and median production remain close (~500 boxes/day), indicating consistent day-to-day operations.
- **Machine Utilization:** Average utilization is ~41.7%, well below installed capacity, confirming under-utilization despite stable production.
- **Variability:** Low standard deviation in production (~11 boxes) and utilization (~0.9%) shows performance is consistently constrained, not random.
- **Revenue & Profit:** Daily revenue (~₹60,000) and gross profit (~₹10,000) remain stable across days and months, indicating growth is limited mainly by capacity utilization rather than cost inefficiency.

UTILIZATION ANALYSIS

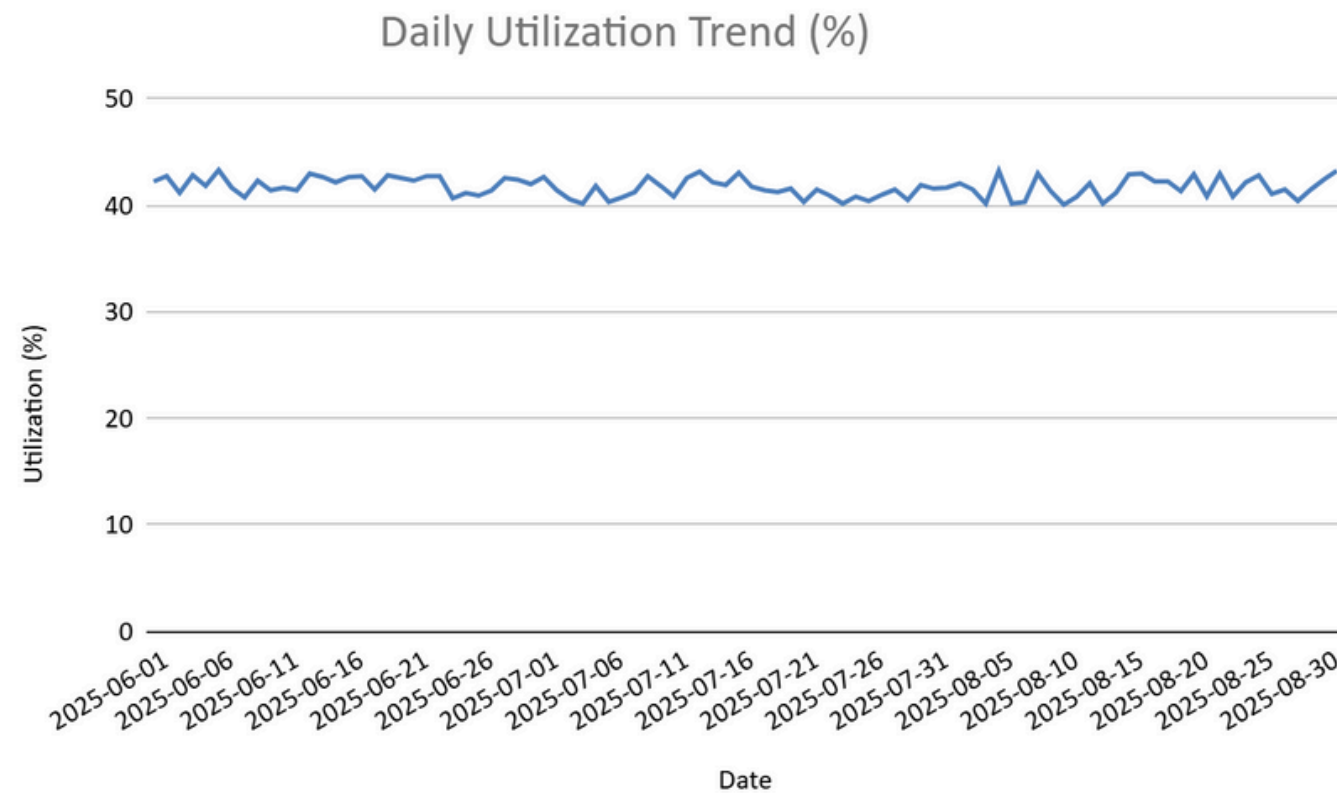


fig 5: daily utilization trend (%)

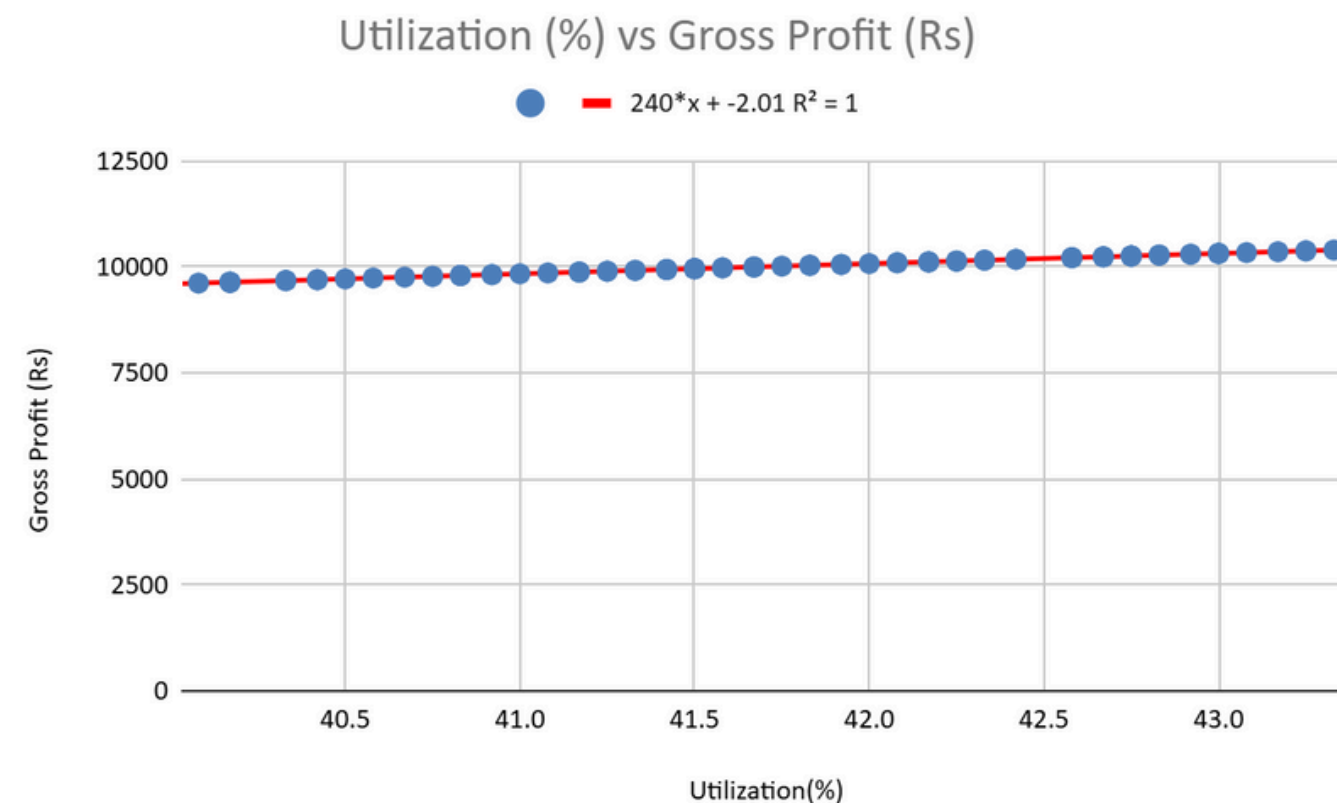


fig 7 : utilization vs gross profit

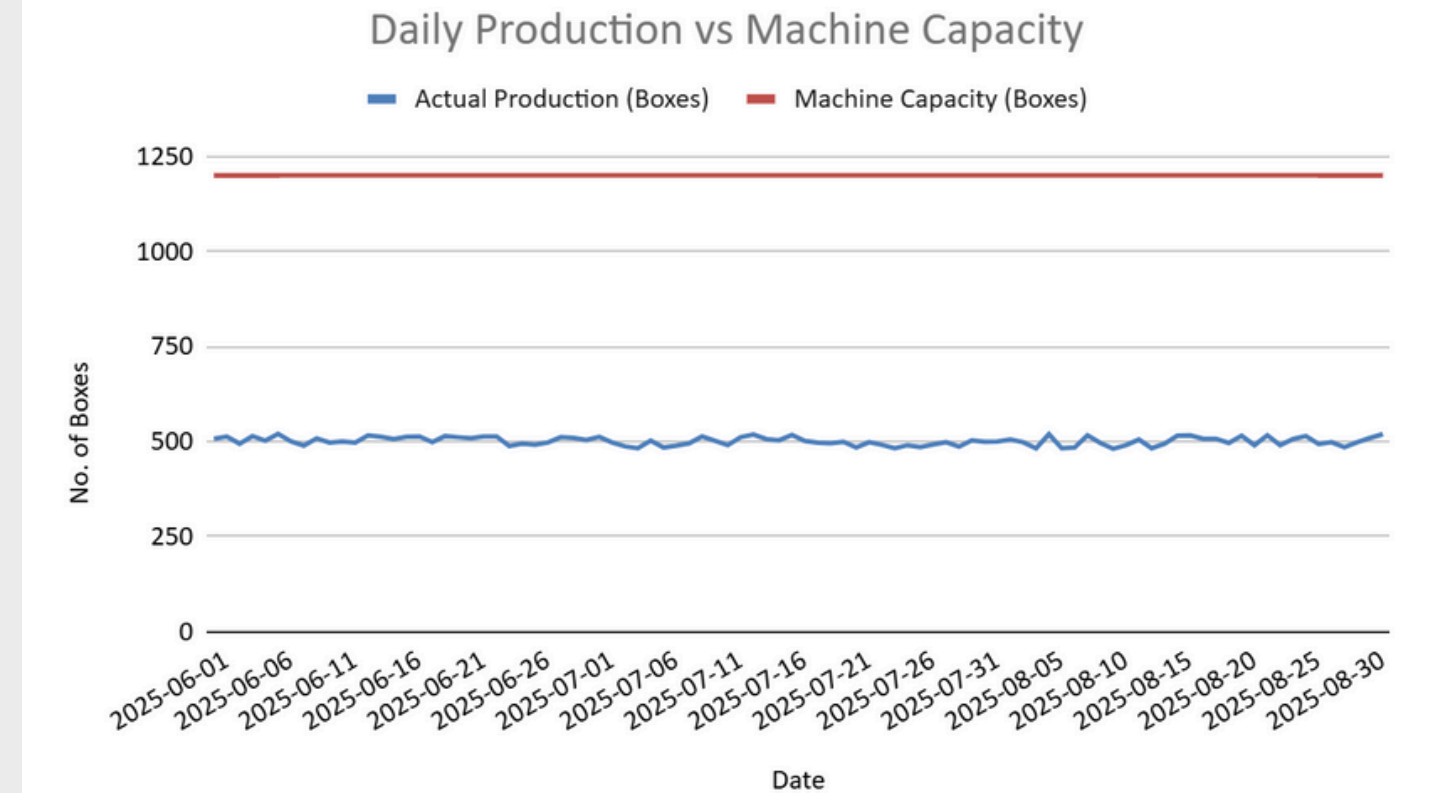


fig 6 : daily production vs machine capacity

Correlation : $r = 0.99$

Observations :

- 1.Utilization-Profit Link: Strong positive correlation observed ($r = 0.99$), indicating higher utilization leads to higher gross profit under a stable cost structure
- 2.Production Constraint: Actual production averages ~500 boxes/day against an installed capacity of 1200 boxes/day, confirming under-utilization
- 3.Revenue Behavior: Revenue and gross profit rise proportionally with utilization, highlighting capacity constraint as the primary bottleneck

Key Formulae Used:

- 1.Utilization (%) = $(\text{Actual Production} / \text{Capacity}) \times 100$
- 2.Revenue (Rs) = Quantity $\times$ Selling Price
- 3.Gross Profit (Rs) = Revenue – Total Cost

CREDIT ANALYSIS

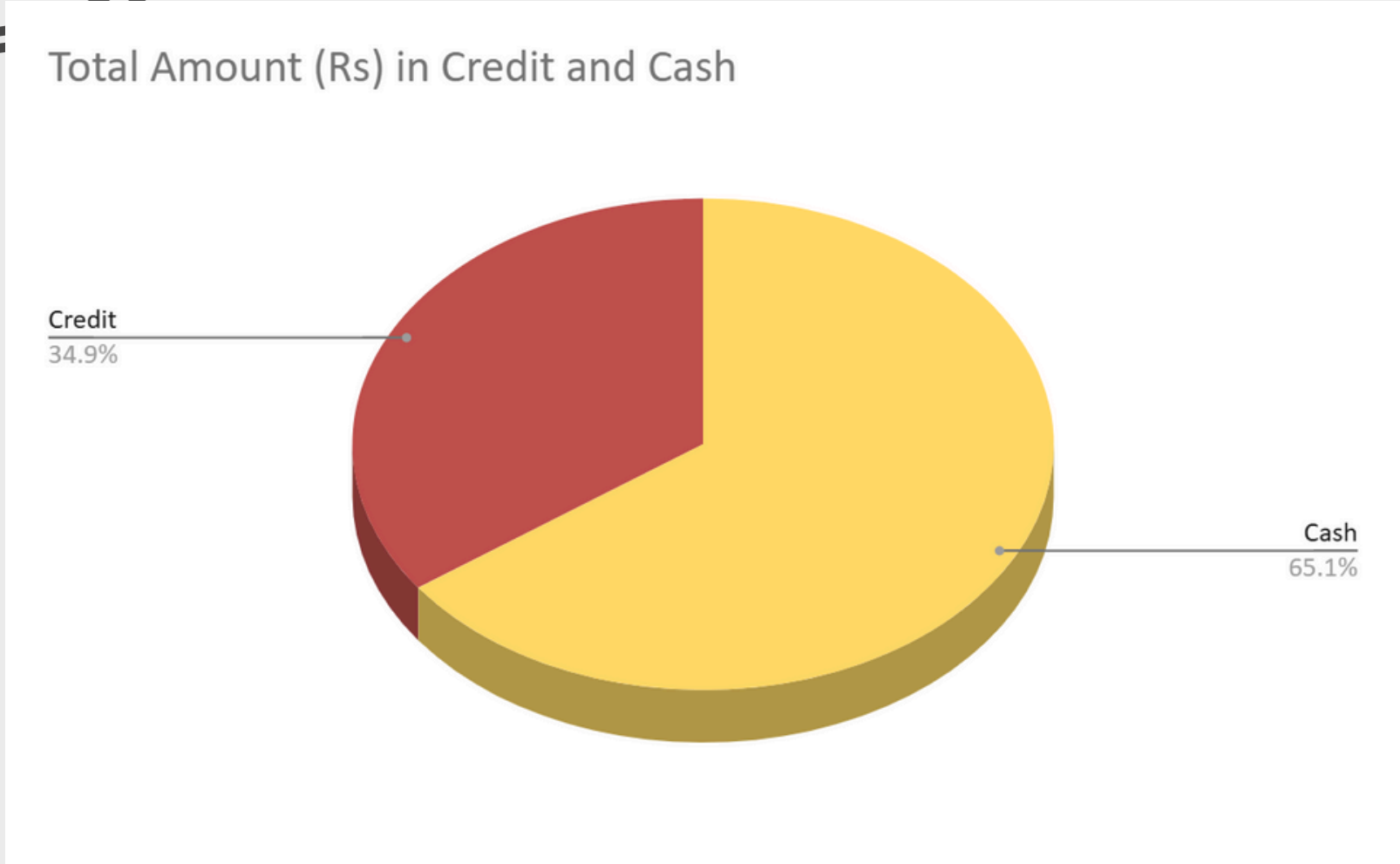


fig 8 : total amount in credit and cash

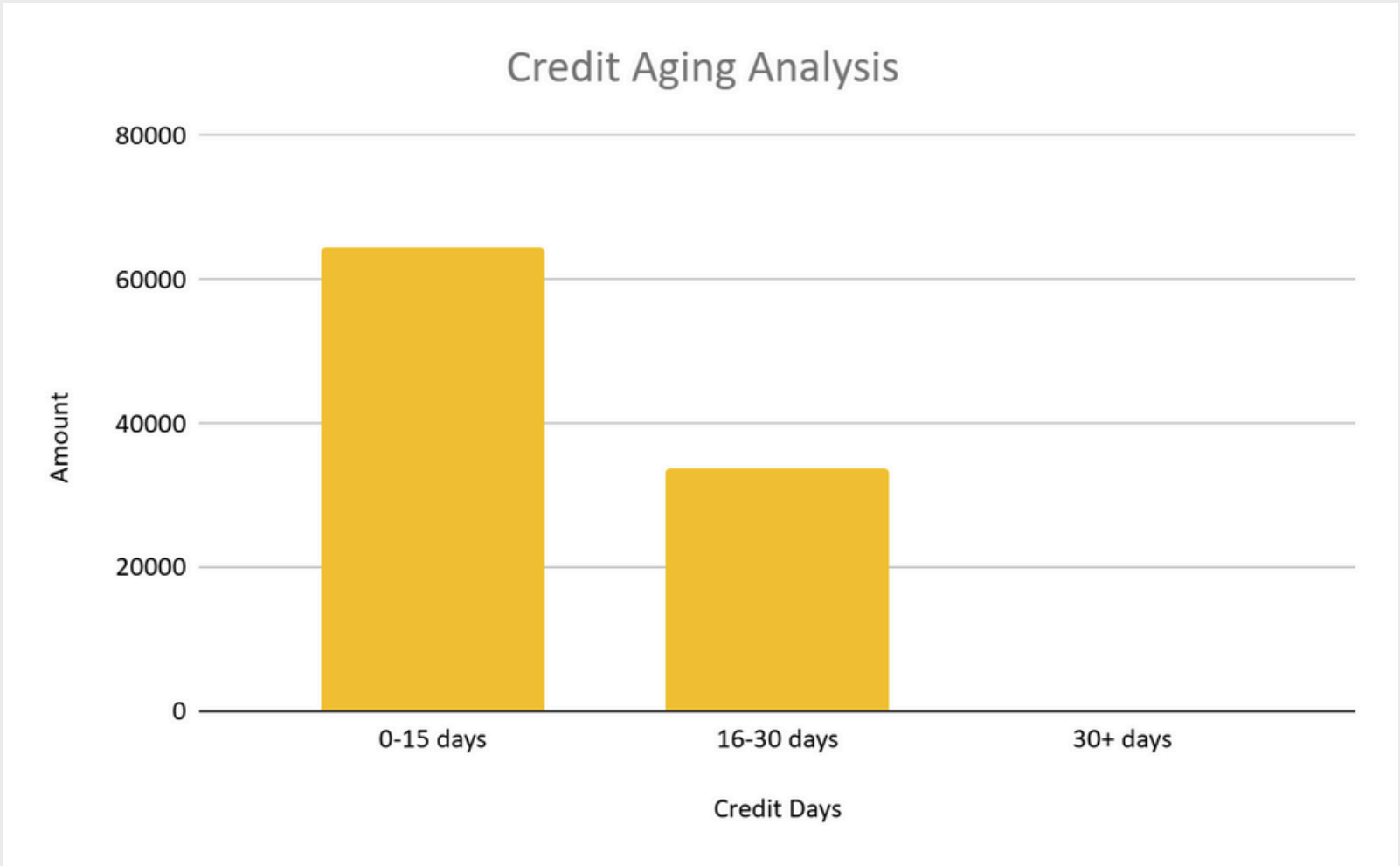


fig 9 : credit aging analysis

- Credit Exposure:** ~35% of total sales are credit-based, indicating moderate dependence on credit
- Cashflow Impact:** Credit sales delay cash inflow compared to cash transactions, creating short-term liquidity pressure
- Credit Pattern:** Most credit is settled within 0-15 days, with some extending to 16-30 days
- Risk Level:** No dues exceed 30 days, indicating controlled credit risk



KEY INSIGHTS

- The business is operationally stable, but growth is structurally constrained by electricity availability rather than inefficiency or demand limitations.
- Machine utilization emerges as the single most critical lever influencing production and profitability.
- Existing infrastructure is underutilized, indicating that meaningful growth is possible without additional machinery investment.
- Credit practices are controlled, but delayed inflows create short-term liquidity pressure, especially relevant if production scales.
- Addressing the root infrastructure constraint can unlock higher output and profitability while maintaining current cost efficiency.
- The business demonstrates cost discipline, as profitability remains stable despite low utilization, indicating that fixed and variable costs are well controlled.



RECOMMENDATIONS & CONCLUSION

Recommendations :

- **Improve Power Reliability:** Invest in backup power solutions to reduce downtime and increase machine utilization toward installed capacity.
- **Transformer Installation:** Secure a dedicated transformer and sanctioned heavy electrical supply to remove the primary production bottleneck and enable utilization of installed capacity.
- **Strengthen Credit Discipline:** Encourage faster collections by prioritizing reliable customers and limiting credit duration where possible.
- **Increase Cash Sales Share:** Offer small incentives for cash payments to improve short-term liquidity and reduce working capital pressure.

Conclusion :

- The business operates with stable and consistent performance, but profitability is structurally constrained by low machine utilization which is backed by limited power supply.
- Credit risk is controlled, yet delayed cash inflows create short-term liquidity challenges.
- Data-driven operational improvements, rather than capacity expansion, can significantly enhance profitability and cashflow sustainability.